AAKRUTI Innovation Competition

AAKRUTI INNOVATION COMPETITION 2026 · INDIA SEMI-FINAL

AquaSentinel

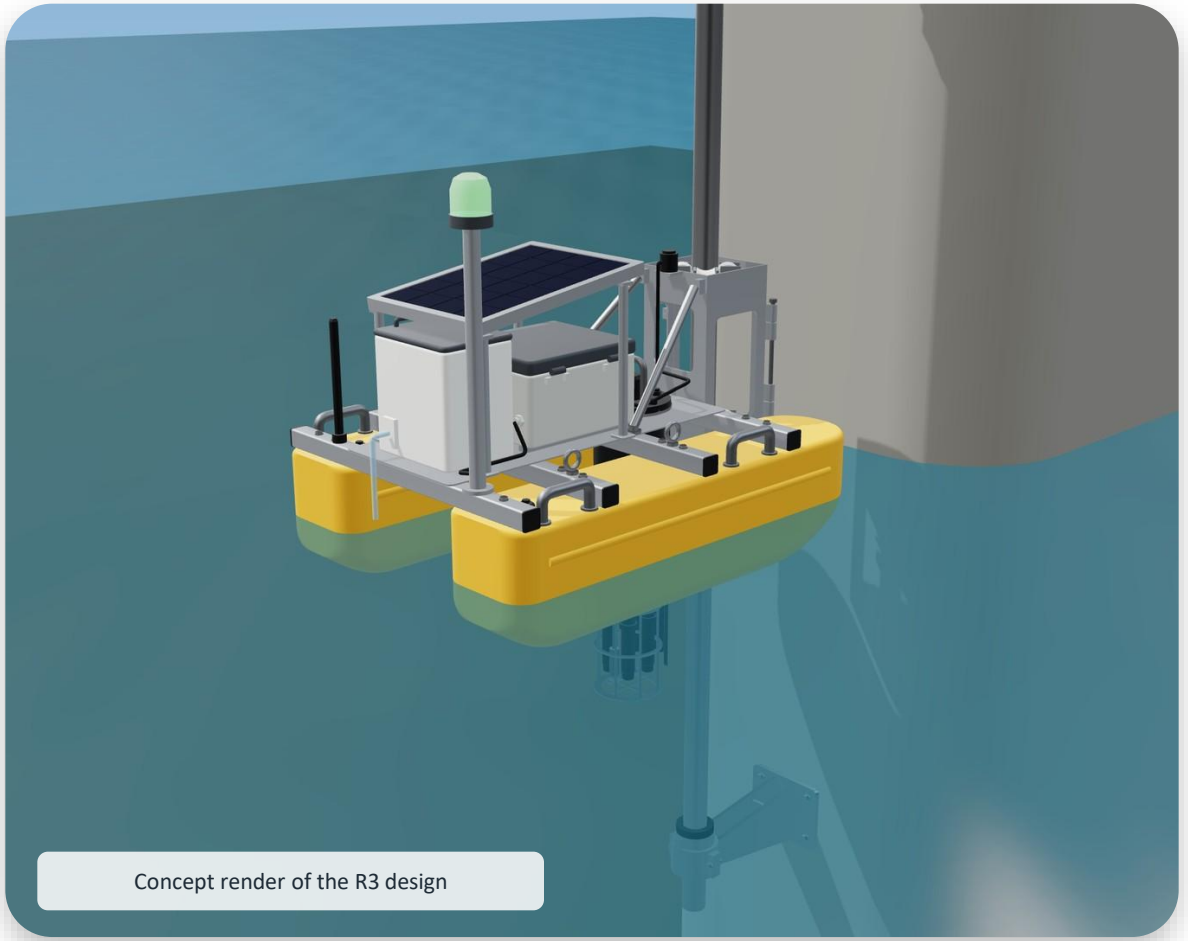
Catches the spill that monthly sampling misses.

An autonomous river robot on a guide pole. It is designed to read the river every 15 minutes, decide on board, and when a spill passes, turn its beacon red, alert the water-works and bottle a sample for the lab.

SENSE every 15 min

THINK on board

ACT beacon, alert, sample



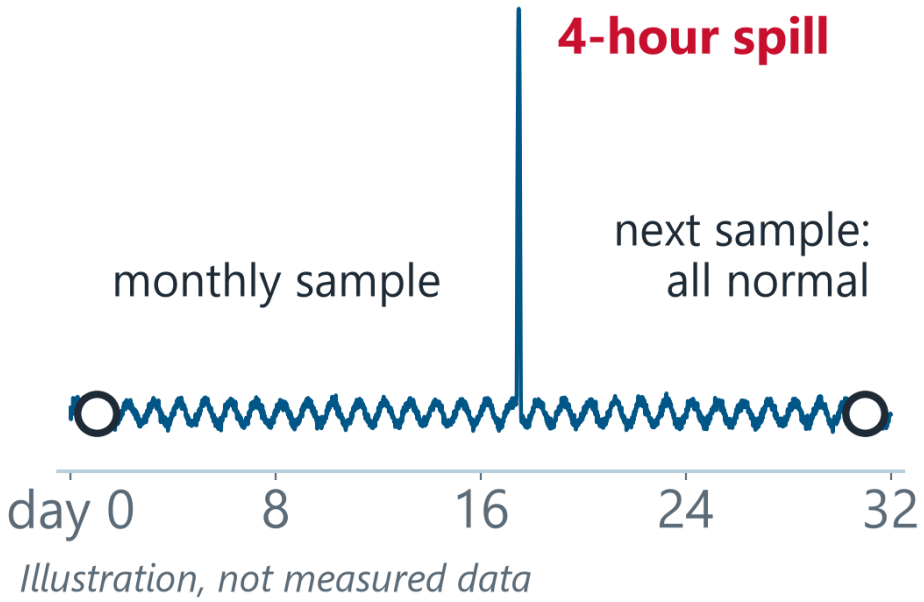
Concept render of the R3 design

Team Sanjivani Hero's Journey
Sanjivani University, Kopargaon · AAKRUTI 2026 · Robotics · AG26-1500$41

A spill passes in hours; sampling is periodic

Monthly checks miss a spill

conductivity at one river point



Illustration, not measured data

54 of 296 — polluted river stretches/locations in India are in Maharashtra, still the most of any state (CPCB, 2025)

Monthly or quarterly — manual grab samples at most national monitoring points (NWMP)

2:00 a.m. — a night-time discharge is gone long before the next sample

54 of 296: CPCB assessment published 2025 (2022–23 data; PIB, 16 Mar 2026).

SOLIDWORKS 3DEXPERIENCE Edu

AquaSentinel in 60 seconds

Marketing video

60 seconds

Click to play



Our own renders and graphics only

Captions burned in: works without sound

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An autonomous river robot: sense, think, act

1 SENSE

pH, conductivity, dissolved oxygen, turbidity, temperature and river level

2 THINK

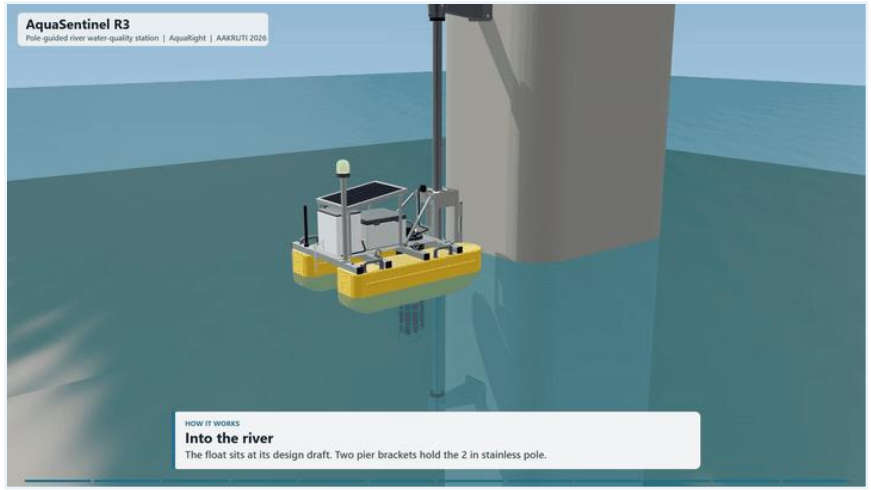
An ESP32 on board decides every 15 minutes, with no cloud needed

3 ACT

Beacon, LTE alert, and a pump, valve and 4-bottle carousel take a sample

Objective	Target	Status
Detect a spill	flagged within 30 min	● shown on synthetic data only
Keep station, no drive motor	ride 2 m of river rise	● calculated · 1:5 model next, then tank test
Survive a flood	1.5 m/s current + 500 N debris	● static case calculated · impact and anchors to verify
Run without sun	about 12 days (with beacon)	● design estimate · measure on the bench
Service from the bridge	no tools, nobody wades in	● designed (tool-free gate)

The river moves it; the robot decides



Our CAD animation: the river lifts the float up its pole

2 m
of river rise

0
drive motors

8
rollers, 2 tiers

1. A 2-inch 316L guide pole clamped to a bridge pier: no riverbed work
2. 8 rollers in 2 tiers let the river lift the float through 2 m
3. Its side area sits behind the pole, so it trails the current like a weathervane
4. A pull-up probe cartridge; a wiper cleans the optical windows every 6 hours

Every 15 minutes, on the float

Hard limit crossed
pH < 6 or > 9 · DO < 4 mg/L · EC > 2250 µS/cm

2 of the last 3 readings unusual
vs a baseline learned for each hour of the day

→ **ACT**
beacon turns red
LTE alert
1 bottle sampled

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Every requirement checked on the CAD model

143 mm

freeboard

requirement ≥ 100 mm

2.27×

reserve buoyancy

requirement ≥ 2.0×

60°

righting arm still positive at

max 180 mm at 25°

642 N

flood + debris load at the float

142 N drag at 1.5 m/s + 500 N debris

2.66

pole safety factor, static load case

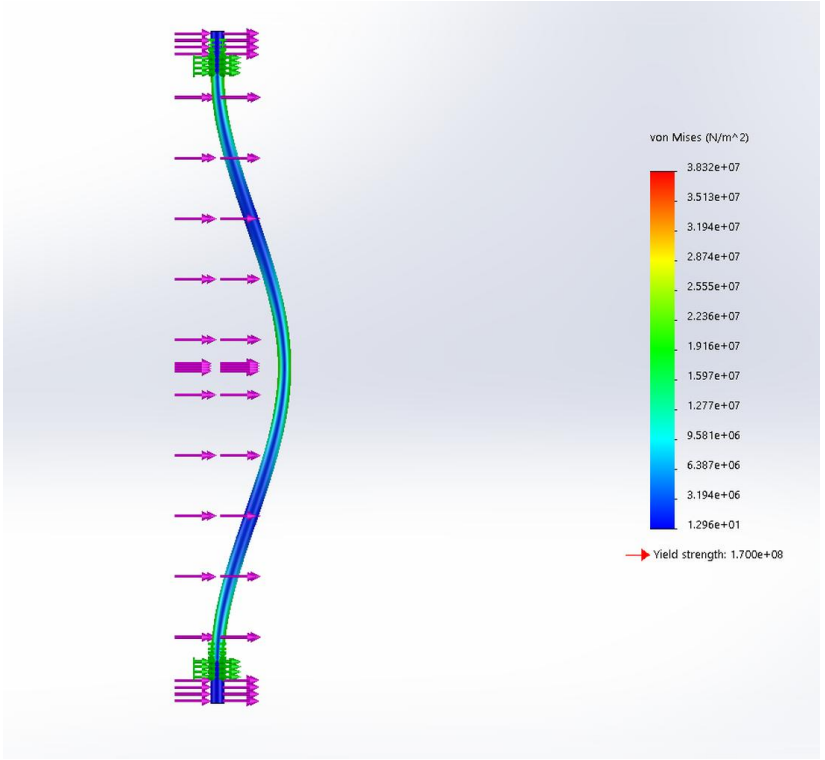
pinned + carriage moment: 64 vs 170 MPa

8.6×

carriage anti-jam margin

1.7× even with seized rollers

14 of 14 design checks pass · values computed from the R3 CAD geometry

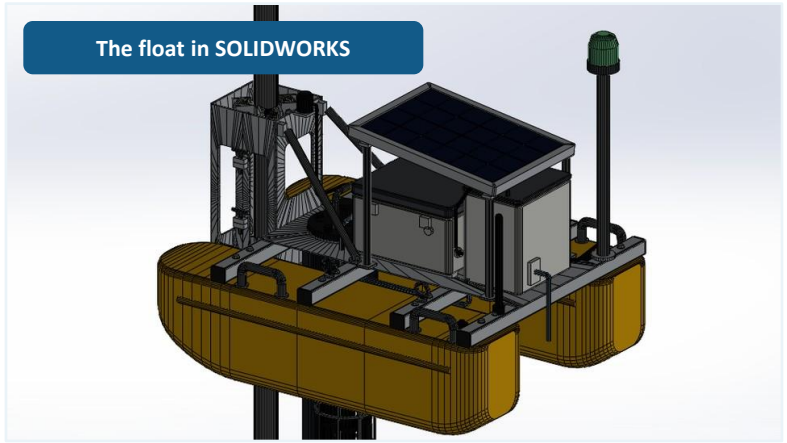


SOLIDWORKS Simulation of the pole under the flood + debris load. Fixed ends give 4.4; we quote 2.66. Static case: dynamic impact and anchors still to verify.

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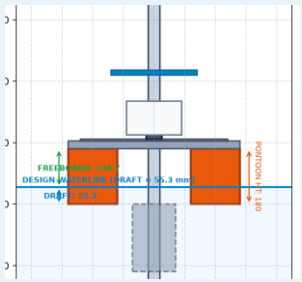
66 parts, one mould, off-the-shelf electronics



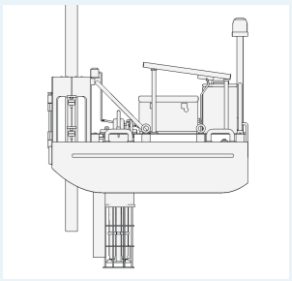


PE hulls	UV-stabilised, rotomoulded: one mould makes both hulls
6061 aluminium	deck and carriage: cut, bent and TIG-welded
316L stainless	pole and brackets: standard 2-inch pipe and 6 mm plate
Off the shelf	RS485 probes, ESP32, LTE modem, 20 W solar, LiFePO4 battery

CAD caught our two biggest mistakes



v2 (old values): pole through the pod



R3: pole at the bow

Real mass from CAD: 25.4 → 48.5 kg, so the hulls grew (draft 55 → 117 mm)

Built with SOLIDWORKS, SOLIDWORKS Simulation, Pack and Go and ENOVIA Project Planner

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A service for those who must act on spills

Pollution boards

catch discharges; know when to sample

Water-works

close the intake in time

Irrigation / WRD

a continuous record with river level

Industrial estates

an independent downstream record

What it costs



≈ ₹1.57 lakh

per float, ex-GST, + ₹0.29 lakh bridge kit

- probes 44 %
- everything else 56 %

≈ ₹1.24 lakh at 100 units (estimate)

Versus a real-time station

₹35–40 lakh

for one station's 10-parameter instruments (CSE)

Not like-for-like: stations measure more. We add the dense early-warning layer between them.

How we earn

Monitoring as a service

- ₹12,000 a month per float, 5-year contract: hardware, install, service, alerts, data
- Hardware paid back in ≈ 20 months; ≈ 46 % margin over 5 years
- Or buy: ≈ ₹3 lakh installed + maintenance contract

Kopargaon pilot → **3 floats, one stretch** → **MPCB regional programme** → **Maharashtra's 54 polluted sites**

Sources: CPCB assessment (2025); CSE (station cost); our costed R3 bill of materials (Indian prices, 23 Oct 2026, ex-GST). Fee (5-year contract) and roll-out are proposals, at the 100-unit cost.

Where we are, and what proves it next

✓ **Designed**
66-part CAD, hand calculations, SOLIDWORKS Simulation

✓ **Simulated**
synthetic-data test only: 4 of 4 injected spills flagged within 30 min

3 **Before the finale: 1:5 model in water**
draft, level trim and the float riding up its pole, measured

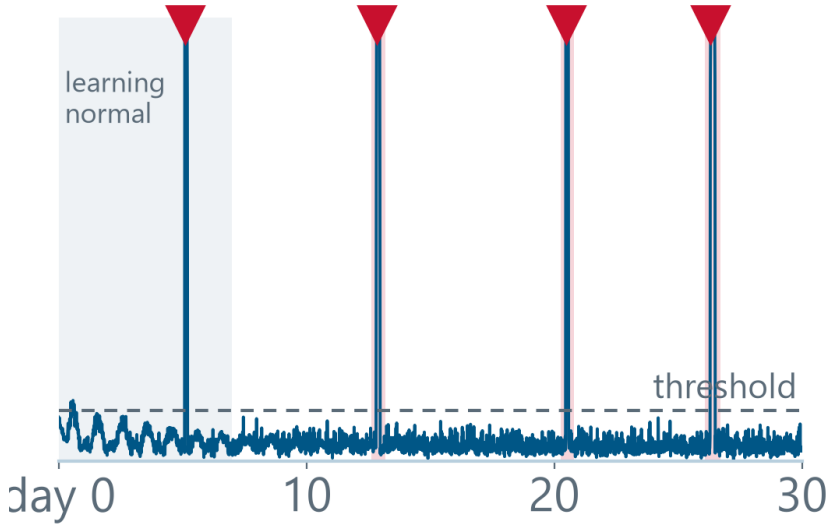
4 **Before the finale: electronics bench demo**
ESP32 with a real probe: reads, decides, acts (light, pump, alert)

5 **After the finale: full-size tank test**
draft, 2 m travel and roller friction, measured

6 **After permits: 30-day Godavari pilot**
probes checked against lab samples at Kopargaon

4 of 4 spills flagged

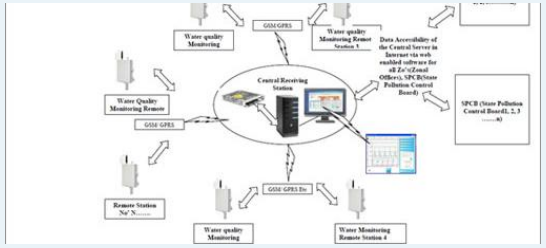
how unusual each reading is (robust z)



Firmware logic on 30 days of SYNTHETIC data

Not tested in water yet

Prior art: what already exists, and what it does well



CPCB real-time stations (Ganga)

Deployed system · India

Does well
- ≈ 17 parameters incl. BOD, COD, ammonia, TOC, level
- Every 15 min over GSM/GPRS to a central server
- 36 stations commissioned in 2017

Trade-offs
- Instruments ≈ ₹35–40 lakh per station, so few stations
- ; Fixed sites physical sampling n.p.v.

Source: CPCB CSE



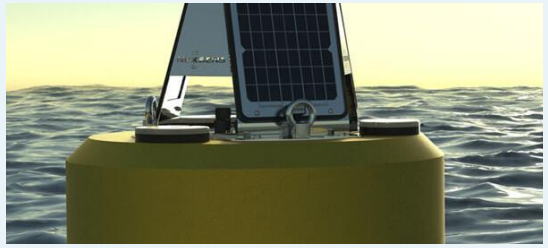
YSI EXO2 sonde (Xylem)

Commercial product

Does well
- 7 ports: up to 6 sensors + anti-fouling wiper
- RS-485, SDI-12, Modbus output
- ≈ 90 days on 4 D-cells at 15-min logging

Trade-offs
- An instrument: needs a mount, logger and telemetry for live alerts
- Price on request

Source: YSI



NexSens CB-450 data buoy

Commercial product

Does well
- Carries sondes and sensors, 3 × 15 W solar
- LTE or Iridium with the X3 logger; web data
- 86 cm foam hull, 56 kg

Trade-offs
- Moored by anchors (3 eye-nuts), not on a guide
- Beacon and sampler not in the listed product; price not listed

Source: NexSens



Teledyne ISCO 6712 sampler

Commercial product

Does well
- From 24 × 1 L bottles to one 5.5 gal container
- Time, flow or event sampling; triggers on up to 16 inputs incl. SDI-12 sondes
- Optional modem dial-out alarms

Trade-offs
- Bank or manhole unit with an intake hose
- Collects samples; no continuous readings itself

Source: RS Hydro / Teledyne ISCO

Patents (preliminary, English-language search): related documents exist, e.g. CN117446093B (floating detection units held by fixed rods) and US8352213 (pier-mounted rod sliding vertically, for riverbed level). No exact match found. Not a novelty or freedom-to-operate opinion.

Product facts from manufacturer or distributor pages, looked up 24 Oct 2026. n.p.v. = not publicly verified. Images: CPCB via CSE, YSI, NexSens, RS Hydro.

Existing solutions vs AquaSentinel (design)

	CPCB real-time station	YSI EXO2 sonde	NexSens CB-450 buoy	ISCO 6712 sampler	AquaSentinel R3 (design)
Readings, parameters	every 15 min; ≈ 17 incl. BOD, COD	user-set; up to 6 sensors	set by its logger; carries sondes	samples only; reads a sonde via SDI-12	every 15 min; pH, EC, DO, turbidity, temp, level
Mounting	fixed station; layout n.p.v.	user-supplied mount, pipe or buoy	moored, 3 eye-nuts	bank or manhole, intake hose	float on a guide pole clamped to a bridge pier
River-level change	n.p.v.	depends on mount	floats within mooring scope	hose reach (max lift per spec)	rides 2 m on 8 rollers, no drive motor (calculated)
Decision on site	data to central server	logs; alarm logic not listed	logger and web portal; alarms n.p.v.	event trigger on up to 16 inputs	on board: limits + learned baseline (synthetic test)
Local light / remote alert	n.p.v.	not part of the sonde	n.p.v.	modem dial-out alarm option	beacon + LTE/SMS alert (designed)
Physical sample	n.p.v.	no (sensor)	not in listed product	yes, up to 24 bottles	4 × 250 mL, one per event (designed)
Indicative cost	₹35–40 lakh instruments (CSE)	price on request	not listed	not listed	≈ ₹1.87 lakh prototype estimate incl. bridge kit
Power, upkeep	calibration, batteries (CSE)	4 D-cells, ≈ 90 days at 15 min	3 × 15 W solar	12 V DC battery or mains	20 W solar + LiFePO4, ≈ 12 days dark (estimate); monthly visit

What we combine differently (narrow, to be proven): each capability exists separately, and a buoy + sonde + triggered sampler could be integrated. Our difference is the architecture: a pier-guided float that follows river level without a mooring, an on-board decision, a beacon for people on the bank and one-bottle-per-event sampling in one unit, at a prototype estimate under ₹2 lakh. Not yet validated in a river; no novelty claim.

Sources as on slide 12: CPCB · CSE · YSI EXO2 · NexSens CB-450 · ISCO 6712 · n.p.v. = not publicly verified. AquaSentinel values are design estimates (backup A1–A5).

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Watching the river every 15 minutes

1. **Earlier warning for people downstream**
2. **A sample that tells the lab where to look**
3. **A light anyone on the bank can read**



Soham Kadu

Project Lead



Shriraj Kamble

System Design



Bhakti Kadam

Electronics Lead



Tanmay Pendbhaje

Software Lead



Dr. Kiran Wakchaure

Mentor

Thank you. We'd love your questions.

Sanjivani Hero's Journey · Sanjivani University, Kopargaon · AG26-1500$41

Questions? Click a topic

A1 All 14 design checks pass

A2 Pole safety factor: we quote the lowest

A3 Power: about 12 days without sun

A4 Decision engine, and the fixes we found

A5 Cost: ₹1.57 lakh, ₹1.24 lakh at 100

A6 Market: a large problem, sparse monitoring

A7 Risks we know, and how we handle them

A8 Roadmap to the finale and the pilot

A9 Prototype levels: where the 1:5 model fits

Explore it live (opens the browser)

3D model ↗ | Dashboard ↗ | GitHub ↗

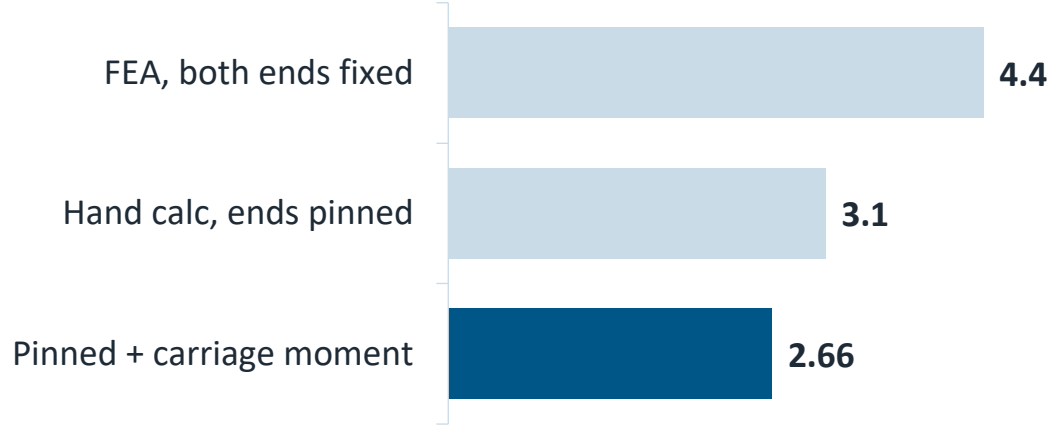
A1 · All 14 design checks pass

Check	Result	Criterion
Reserve buoyancy	2.27×	≥ 2.0×
Freeboard	143 mm	≥ 100 mm
Transverse GM	0.55 m	≥ 0.15 m (IMO reference)
Righting arm	positive to 60°, max 180 mm at 25°	positive to 60°
Pole safety factor, pinned	3.1 (2.66 with carriage moment)	≥ 2.0
Pole deflection, pinned	5.8 mm	≤ span / 200 (13.4 mm)
Jamming, rollers (μ 0.05)	margin 8.6	no self-locking
Jamming, seized rollers (μ 0.25)	margin 1.7	no self-locking
Float lag at full design load	17.7 mm	≤ 30 mm
Weathervane stability	lateral centre 351 mm aft of the pole	aft of the pole
Pole corridor clearance	9.7 mm	≥ 5 mm
Float vs fixed parts, 2 m travel	no clash	none
Only rub rails touch the pier	rub rail 12 mm outboard	rub rails outermost
Part-to-part interference	none in 101 pairs	none

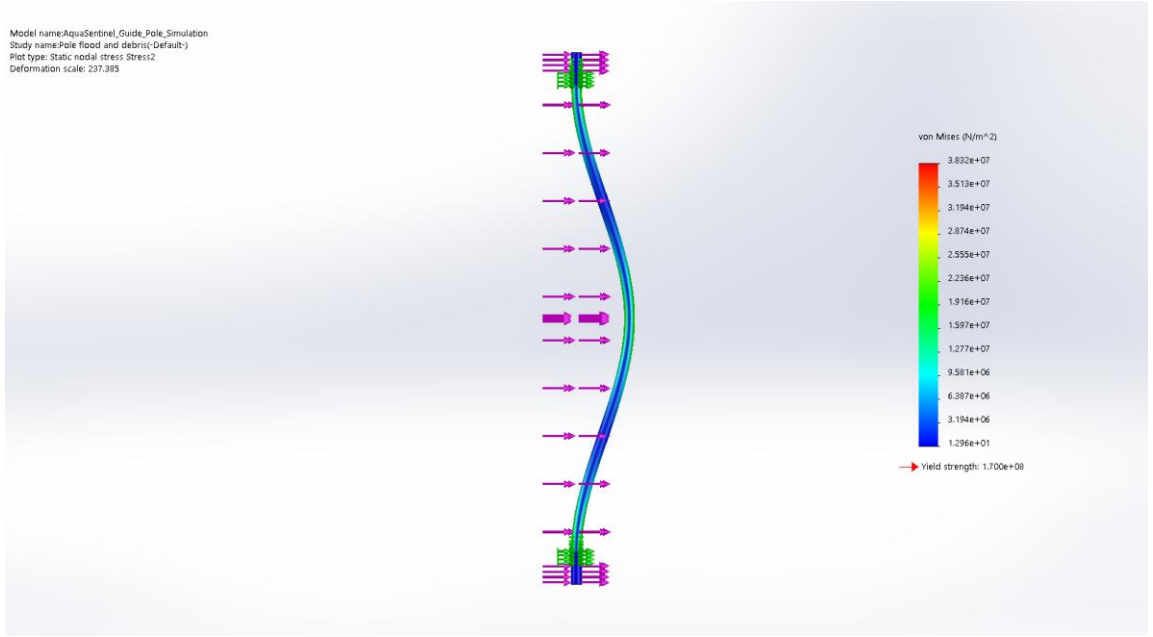
Source: mechanical/checks/design_checks.json, computed from the R3 CAD; masses partly catalogue estimates.

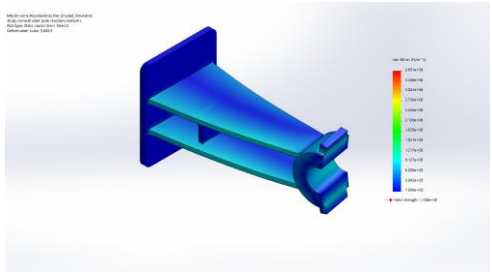
A2 · Pole safety factor: we quote the lowest

Pole safety factor by idealisation (requirement ≥ 2.0)



- Peak stress in MPa: FEA 38.3 · pinned 54.8 · pinned + moment 64.0 (316L yield 170)
- Load: 642 N at the float = drag ½·1000·1.5²·1.05·0.120 m² = 142 N + 500 N debris (quasi-static), plus drag on the pole
- Not yet included: debris impact factor, fatigue, vortex shedding

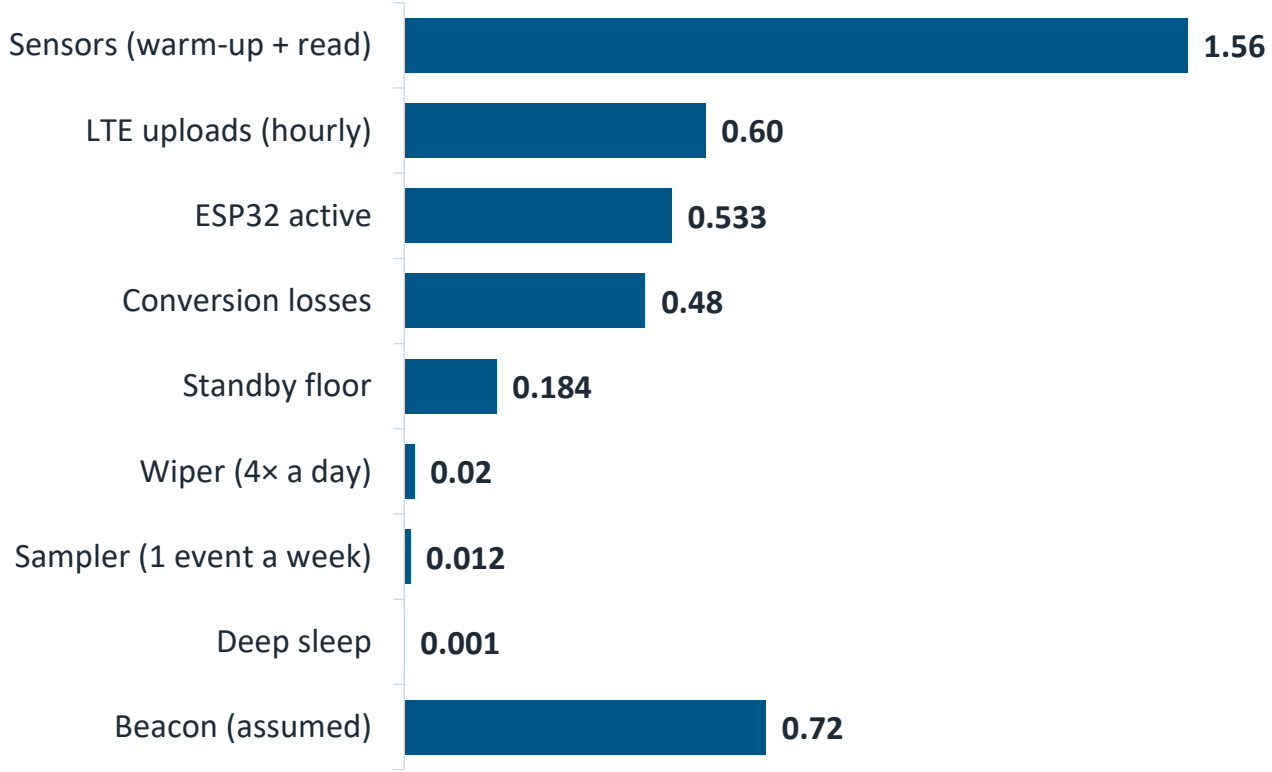




Bracket: 3.7 MPa peak under the 430 N reaction. Anchors, welds and debris impact still to check.

A3 · Power: about 12 days without sun (estimate)

Daily energy by consumer, Wh per day (design estimate)



4.11 Wh/day

nominal, beacon included; 4.93 with a 20 % margin

61.4 Wh

usable: 12.8 V 6 Ah LiFePO4 at 80 % depth of discharge

≈ 12 days

with no sun (preliminary; 15 without the beacon)

39 Wh/day

20 W × 3.0 sun-hours on a monsoon day, after 35 % losses

Beacon assumed: 3 W strobe, 2 % duty, 12 h a night = 0.72 Wh/day. Pump, LTE peaks: measure on the bench

A4 · Decision engine, and the fixes we found

How it decides (designed)

- Reads pH, EC, DO and turbidity every 15 minutes; temperature and level are logged
- Hard limits act at once, also while learning: pH < 6 or > 9, DO < 4 mg/L (CPCB Class C/D minimum), EC > 2,250 µS/cm (CPCB Class E maximum); our operating limits, tunable per site
- A baseline per hour of the day (EWMA); a robust z-score above 3 counts as unusual
- EVENT when 2 of the last 3 readings are unusual; normal again after 4 clean readings
- First 7 days: baseline learning only. An early-warning system, not a chemical identifier: the lab confirms

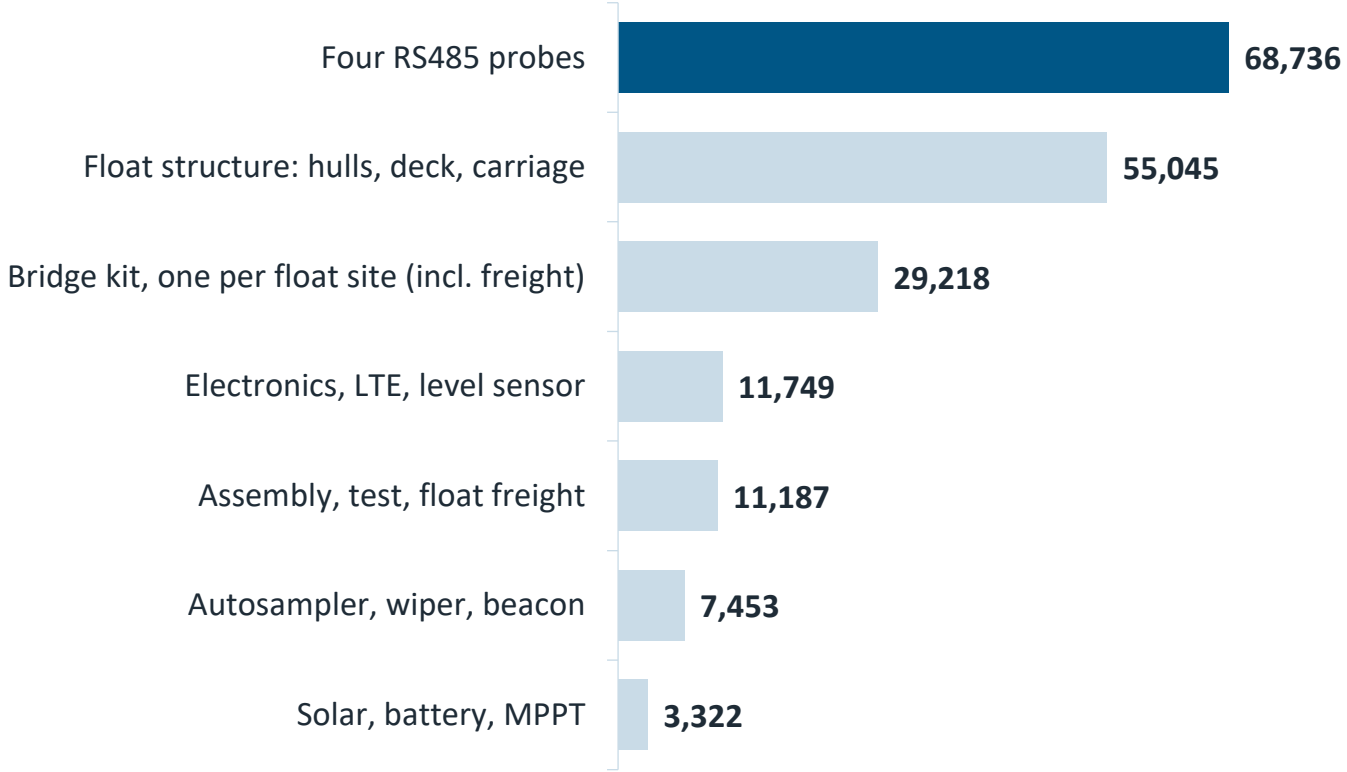
Our audit's fixes: in the firmware, bench test next

- A failed probe read sets a sensor-fault state, never 'normal water'
- Save the learned baseline to flash, so a reboot doesn't erase it
- One bottle per event, so 4 bottles cover 4 separate events
- Clear stale alarm state when learning ends
- Use river level to tell monsoon runoff from a discharge (planned)

Sources: our audit's analytics review (18 Oct); firmware anomaly_engine.cpp; analytics/evaluate.py on the synthetic dataset.

A5 · Cost: ₹1.57 lakh a float, ₹1.24 lakh at 100

Prototype cost: one float + its bridge kit, ₹ ex-GST (researched 23 Oct)



≈ ₹1.57 lakh

per float, ex-GST; bridge kit ₹0.29 lakh at each float's bridge; all 66 part numbers priced

–₹32.6k

revised estimate: contingency moved to the pilot budget (₹19.8k) + 4 design simplifications (₹12.8k)

≈ ₹1.24 lakh

per float at 100 units: one rotomould, welding jigs, probe volume price

Cost-down

next: probe volume pricing, a K = 1 EC cell; a Lite float (pH, EC) ≈ ₹1.13 lakh

A6 · Market: a large problem, sparse monitoring

Fact	Number	Source
Polluted river stretches/locations, India	296 on 271 rivers: 156 stretches + 140 locations (CPCB, published 2025, 2022–23 data)	PIB, 16 Mar 2026; CPCB
Maharashtra	54, still the most of any state (55 in the 2022 assessment)	PIB; CPCB
National monitoring network	≈ 4,922 locations, 2,265 on rivers; manual samples, monthly or quarterly	CPCB reply to NGT, OA 413/2025
Real-time station cost	₹35–40 lakh for 10-parameter instruments; ≈ ₹50–84 lakh with systems	CSE; Lok Sabha reply 5115

Illustrative market size (assumptions)

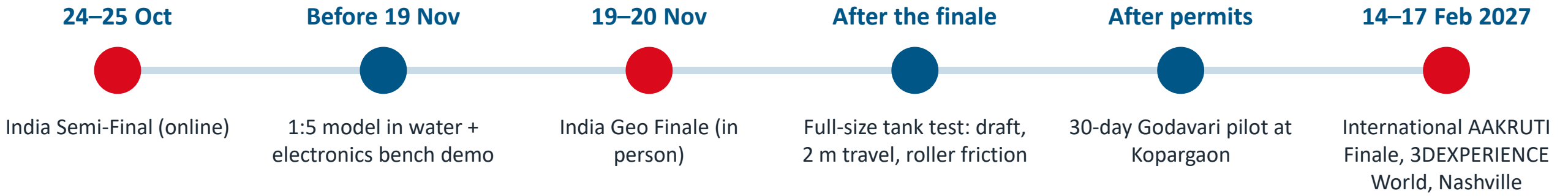
Scope	Floats (3 per counted site)	Revenue at ₹1.44 lakh per float-year
Maharashtra, 54 sites	162	≈ ₹2.33 crore a year
India, 296 sites	888	≈ ₹12.8 crore a year

Market size is illustrative: floats per stretch and the fee (₹12,000 a month) are assumptions to validate in the pilot.

A7 · Risks we know, and how we handle them

Risk	Why it matters	How we handle it
Flood above 2 m of travel	the float is held under at the top stop	get gauge data (CWC/WRD); longer travel or a breakaway link; lift out on flood warnings
Bridge-pier wake	unsteady yaw and cyclic loads	observe in the tank and on site; side-face mounting if needed
Probe fouling and drift	false or missed alarms	wiper; monthly service with before/after readings; lab comparison
Level sensor blind zone	no river level near high water	compare a pressure transducer and a downward radar
Sample is not legal evidence	Water Act §21 needs the Board's procedure	a triage sample that triggers the Board's formal sampling
Beacon colours	red and green read as navigation marks	ask the waterway authority; yellow or shielded light
Permits	no pilot without them	bridge owner, river authority and MPCB: start now

A8 · Roadmap to the finale and the pilot



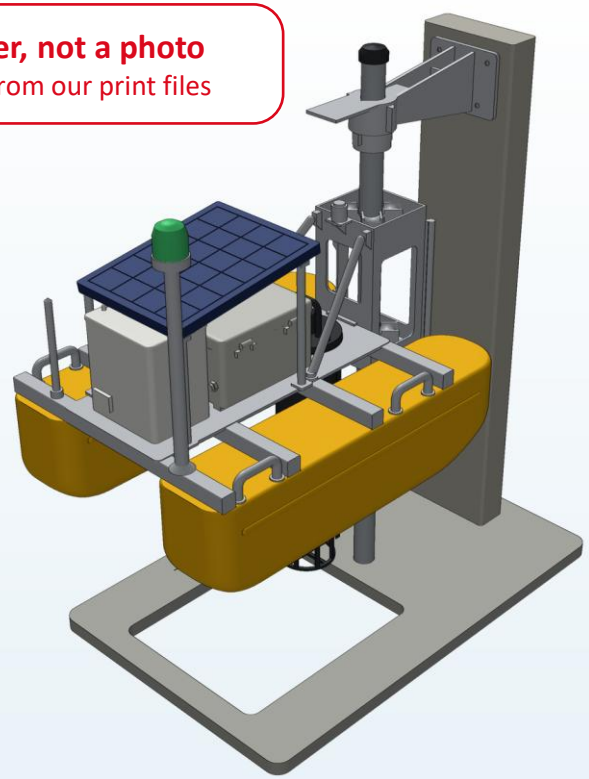
● competition dates (organisers) ● our plan

What we aim to show at the 19–20 Nov finale (our plan)

- Our 1:5 scale model in water: measured draft, level trim, and the float riding up its pole
- The decision logic running on the ESP32 with a real probe, on the bench
- The fixes from our audit: sensor-fault state, saved baseline, one bottle per event

A9 · Prototype levels: where the 1:5 model fits

Render, not a photo
made from our print files



1:5 model of R3 on a pier-and-pole stand: hulls 200 mm, pole 12 mm, 65 mm travel (325 mm full scale)

Level	What it shows	Status
0 · Digital design	CAD, calculations and FEA; logic flags 4 of 4 simulated spills	Done
1 · 1:5 scale model	Layout and mechanism: float slides 65 mm up its pole (not the full 2 m); cartridge lifts out	In progress
2 · Electronics bench demo	ESP32 and a real probe: read, decide, act (light, pump, alert)	Next
3 · Tank test	Full-size float on a 2-inch pole: draft, 2 m travel, roller friction	Planned
4 · Godavari pilot	30 days at Kopargaon; alerts checked against lab samples	After permits

A scale model shows form and mechanism.

It is not a working prototype: there are no electronics inside.